

Where Next? Predicting the Next Unaffordable U.S. City Using Housing & Economic Indicators

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Abstract

This project develops an early-warning system to identify which mid-sized U.S. metropolitan areas are on trajectory toward housing affordability collapse before mass displacement begins. Using training data from three cities that have already experienced such collapse - Austin, TX; Boise, ID, and Tampa, FL. The project engineers temporal housing and economic features, trains on XGBoost binary classifier, and applied the resulting model to score 371 additional metro areas. Feature importance is explained via SHAP values. Results confirm the price-to-income 5 year change is the dominant early-warning signal, followed by affordability momentum and HPI year-over-year growth. The explanatory model (Model 1) serves as the project's primary answer to the research question, The model achieves 90.7% accuracy and a 0.943 ROC-AUC on the explanatory model.

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1. Introduction & Problem Statement

Housing affordability in American cities has deteriorated sharply over the past decade. Cities like Austin, Boise, and Tampa were once affordable, mid-sized metros. Yet, they underwent rapid, and multi-year affordability collapse driven by population influx, constrained housing supply, and surging home values. Working- and middle-class residents were displaced as housing costs outpaced income growth.

Despite the severity of this trend, no reliable early-warning system exists to flag which cities are *next* to tip into unaffordability before displacement is already underway. Policy interventions like zoning reform, housing vouchers, inclusionary requirements, years of lead time. City planners, regional lenders, and housing advocates are currently operating reactively, not proactively.

Primary Research Question

Which socioeconomic and housing market indicators best predict the onset of affordability collapse in mid-sized U.S. metros, and can those indicators identify at-risk cities before displacement begins?

Data Scheme

To allow peers to trace this project's logic from raw source CSVs to final outputs without reconstructing it from the notebook cells, the pipeline is summarized below:

- Merged dataset: 83,699 rows x 25 columns, quarterly observations across all U.S. CBSAs, 2010-2024, stores at data/final/merged_full.csv
- Collapse definition: `is_unaffordable = price_to_income_ratio >= 5.0`; `collapse_onset` = first quarter `is_unaffordable` flips from False to Positive after a period of False.
- Train/test logic (Model 1): Rows before each city's first `collapse_onset` event become the training pool; rows at or after onset become the validation pool. A stratified 70/30 random split (with fixed `random_state = 42`) is then applied within the eligible pool.
- Train/test logic (Model 2): Identical onset-based splitting logic, restricted to Austin and Boise only.
- Holdout set: All 371 non-training CBSAs with completed data for the 3 sourcing features (`hpi_yoy_lag`, `pop_velocity_lag`, `pop_acceleration_lag`), no grounded-truth labels available

2. Literature Review & Context

The Affordability Crisis in Context

Research by the Urban Institute, HUD, and Zillow consistently identifies the price-to-income ratio (PTI) as one of the most reliable cross-market metrics for housing stress. A household spending more than 30% of gross income on housing is considered “cost-burdened” (HUD standard); a home priced at more than 3x median household income is historically considered unaffordable by conventional mortgage underwriting standards.

The 5.0x PTI threshold used in this project aligns with academic and industry literature showing that sustained home prices at or above 5x median income consistently preceded displacement pressure. Research from the National Association of Realtors and the Federal Reserve Bank of Atlanta identifies price-to-income ratios above 5.0 as structural affordability stress indicators, particularly in metros with limited housing supply elasticity.

HPI as a Leading Indicator

The FHFA House Price Index (HPI) used extensively in this project is one of the most reliable and geographically consistent measures of residential price appreciation. Studies by Case-Shiller and the Federal Reserve have demonstrated that sustained 3-year HPI growth above 25-30% signals market overheating. This project’s threshold of 30% over 3 years (`hpi_3yr_growth >= 0.30`) is grounded in this empirical literature.

Momentum-Based Detection

Shiller’s work on “irrational exuberance” and momentum pricing in real estate markets provides theoretical justification for the sustained-momentum condition used here. A single quarter of high YoY (year-over-year) growth is insufficient to identify structural change; consistent growth above 10% for 3+ quarters signal a regime shift, not transient fluctuation.

3. Data Sources & Warehousing

All data is joined at the CBSA (Core-Based Statistical Area) level using CBSA codes at the primary key.

Source	Variable(s)	Granularity
Zillow Research (ZHVI)	Median home values	Monthly -> Quarterly
FHFA House Price Index	Price appreciation index	Quarterly
U.S. Census ACS (B19013_001)	Median household income	Annual

Census Population Estimates	Metro population	Annual
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Tampa CBSA Note: Tampa’s CBSA code requires special handling. The official Census ACS GEOID for Tampa-ST.Petersburg-Clearwater metro is 45300, but the ZHVI data uses 45294 (Tampa MSAD). A crosswalk correction was applied in R:

```
None
income_clean_tampa_fix <- income %>%
  filter(cbsa = 45300) %>%
  mutate(cbsa = 45294) %>%
income_clean <- bind_rows(income_clean, income_clean_tampa_fix)
```

The merged dataset contained **83,699 rows x 25 columns** spanning quarterly observations across all U.S. CBSAs from 2010-2024. The three training cities (Austin: CBSA 12420, Boise: 14260, Tampa: 45294) are explicitly verified in the merged dataset before proceeding with feature engineering.

4. Affordability Collapse Definition & Threshold Selection

Before feature engineering, a precise, operationalizable definition of “affordability collapse onset” was required. Three conditions were tested and adopted:

Condition 1 - Price Acceleration

3-year HPI growth $\geq$ 30% (`hpi_3yr_growth` $\geq$ 0.30)

This captures rapid price escalation. Academic literature (Malpezzi & Wachter, 2006; Case & Shiller, 1989) documents that 3-year real HPI growth above 25-35% is strongly associated with marketing overheating and subsequent affordability deterioration.

Condition 2 - Sustained Momentum

YoY ZHVI growth $\geq$ 10% for 3+ consecutive quarters (`zhvi_yoy` $\geq$ 0.10)

A single-quarter spike may be noise; 3 consecutive quarters of $\geq$ 10% YoY ZHVI growth signals a durable trend. The 10% threshold aligns with Federal Reserve research characterizing “rapid” home price growth.

Condition 3 - Affordability Burden (Primary Label Criterion)

Median home price > 5.0x median household income (price_to_income_ratio >= 5.0)

The PTI threshold of 5.0s was empirically selected after testing four alternatives (4.0, 4.5, 5.0, 5.5) against all three training cities. The threshold sensitivity analysis, conducted in Python, yielded the following onset dates:

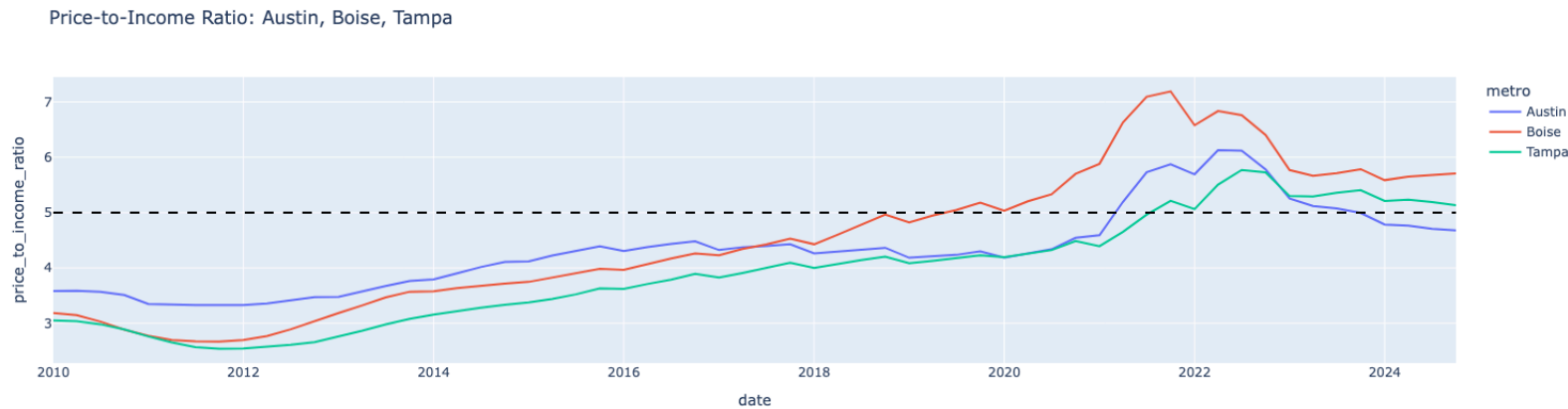
Table 1: Affordability Collapse Onset Dates by PTI Threshold

Threshold	Austin Onset	Boise Onset	Tampa Onset
4.0	2014 Q3	2016 Q2	2017 Q4
4.5	2020 Q4	2017 Q4	2021 Q2
5.0	2021 Q2	2019 Q3	2021 Q4
5.5	2021 Q3	2020 Q4	2022 Q2

5. Exploratory Data Analysis

Figure 1: Price-to-Income Ratio Trajectories (2010-2024) - Austin, Boise, Tampa

Generate in Python using plotly. The plot displays quarterly PTi for all three training cities from 2010 through 2024. A red dashed horizontal line marks the 5.0 affordability collapse threshold.



Key Findings: All three training cities crossed 5.0x PTI threshold during 2019-2021, with Boise the earliest and Austin/Tampa crossing nearly simultaneously during the COIVD-era housing surge.

YoY ZHVI Growth Trends

YoY ZHVI growth for all three cities shows a pronounced acceleration beginning in 2020, consistent with COVID-19-era migration patterns, remote work adoption, and historically low mortgage rates driving demand well ahead of supply.

Population Velocity

Population velocity (year-over-year change in annual population growth) is available for Austin and Boise, confirming accelerating in-migration in the 2-3 years preceding PTI collapse. Tampa does not have population data in this pipeline. This is an acknowledged limitation (see Section 8).

EDA-To-Feature Engineering Bridge

The steep post-2020 acceleration visible in the YoY ZHVI growth trends above directly motivated the 3-quarter sustained-momentum condition (Condition 2) used in the collapse definition. A single-quarter spike, as seen sporadically in pre-2019 data for all three cities, does not appear in Figure 1's PTI trajectories as a lasting threshold crossing, whereas the sustained 2020-2021 climb does. This visual pattern, more than the numeric thresholds alone, justified excluding transient outliers from the `collapse_onset` label.

6. Feature Engineering

All features are lagged by 4 quarters (1 year) before any rolling computation. This is a critical data integrity decision: it ensures no feature accidentally incorporates same-quarter or future information that overlaps with the construction of the `collapse_onset` label, preventing data leakage.

Design Rule: *Never let a feature use current or future information that overlaps with label construction.*

Features Engineered

Feature 1: Price-to-Income 5-Year Change (`price_to_income_5yr_chg`)

The change in PTI from 20 quarters (5 years) prior, computed on the lagged PTI series. This captures structural, slow-moving affordability deterioration.

Python

```
df['price_to_income_lag'] = g['price_to_income_ratio'].shift(LAG)
df['price_to_income_5yr_chg'] =
df.groupby('cbsa')['price_to_income_lag'].transform(lambda x: x - x.shift(20))
)
```


Feature 2: Affordability Momentum (affordability_momentum)

A 12-quarter rolling linear slope of lagged YoY ZHVI growth. This is a smooth, directional signal capturing whether price growth is accelerating or decelerating over a 3-year window.

Python

```
df['affordability_momentum'] = df.groupby('cbsa')['zhvi_yoy_lag'].transform(lambda
x: x.rolling(12, min_periods=12).apply(
    lambda y: np.polyfit(np.arange(len(y)), y, 1) [0] if y.notna().all() else
    np.nan
))
```

Feature 3: HPI YoY Lag (hpi_yoy_lag)

Year-over-year change in the FHFA House Price Index, lagged 1 year. This is the most broadly available feature and the only one with multi-city holdout coverage (66,325 rows across 407 cities).

Feature 4&5: Population Velocity & Acceleration (pop_velocity_lag, pop_acceleration_lag)

First and second difference of annual population YoY growth, lagged 1 year. Available for Austin and Boise but not for **Tampa**. This is due to a known data gap. Used exclusively in model 2 (generalization/scoring model) and holdout scoring.

Table 2: Feature Coverage by City

Feature	Austin (non-null)	Boise (non-null)	Tampa (non-null)
affordability_momentum	85	81	145
zhvi_qoq_lag	99	95	159
price_to_income_5yr_chg	40	40	100
hpi_yoy_lag	187	179	244
pop_velocity_lag	55	55	0
pop_acceleration_lag	54	54	0

Tampa's population feature gap results in pop_velocity_lag = 0, requiring separate model architectures for explanatory vs. generalization use cases.

7. Predictive Modeling

Two models were developed with distinct purposes. Model 1 is designated as this project's primary model for answering the research question, since it uses the strongest available feature set and achieves the highest predictive accuracy. Model 2 exists specially to extend scoring coverage to cities lacking Model 1's full feature set, and its lower performance is an accepted trade-off for that broader reach, not completing claim to primacy.

Model 1 - Explanatory model (All 3 Training Cities, 4 Features)

Purpose: Understand *which features drive collapse* and generate SHAP interpretability; **serves as this project's primary answer to the research question.**

Target Variable: is_unaffordable (binary: PTI>5.0)

Features: affordability_momentum, zhvi_qoq_lag, price_to_income_5yr_chg, hpi_yoy_lag

Training Cities: Austin, Boise, Tampa

Split: 70/30 stratified random split (train : {0: 85, 1:41}; val: {0: 37, 1:17})

Architecture: XGBoost classifier with shallow depth to prevent overfitting on the limited sample:

Python

```
model1_fresh = xgb.XGBClassifier( n_estimators=100, max_depth=3,  
learning_rate=0.05, subsample=0.8, colsample_bytree=0.8, eval_metric="logloss",  
random_state=42)
```

Table 3: Model 1 Performance (Validation Set)

Metric	Score
Accuracy	90.7%
Precision	83.3%
Recall	88.2%
F1-Score	85.7%
ROC-AUC	0.943

Classification Report:

Python

	precision	recall	f1-score	support
0	0.94	0.92	0.93	37

1	0.83	0.88	0.86	17
accuracy			0.91	54

SHAP Feature Importance - Model 1

SHAP (Shapely Additive exPlanations) values were computed using `shape.TreeExplainer` on the training set.

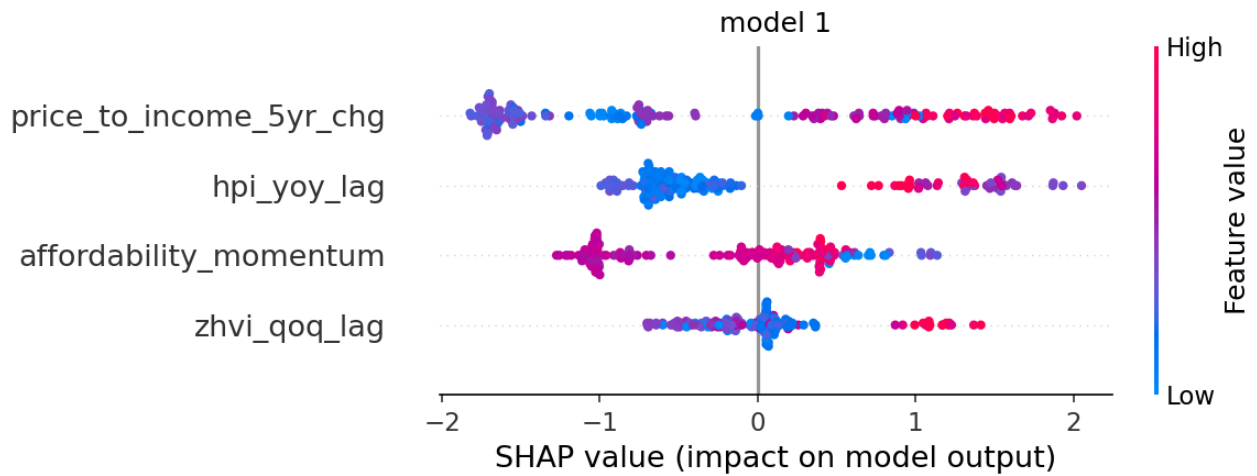


Figure 2: SHAP Summary Plot - Model 1

This is a beeswarm plot that shows the distribution of SHAP values for each feature across all training observations. Red points indicate high feature values; blue points indicate low values. Rightward displacement indicates a positive impact on classifying a quarter as “unaffordable”.

Table 4: Mean Absolute SHAP Values - Model 1

Feature	Mean
price_to_income_5yr_chg	1.083
affordability_momentum	0.702
hpi_yoy_lag	0.588

zhvi_qoq_lag	0.138
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Analysis: The 5-year change in price-to-income ratio is by far the dominant feature, contributing more than the other three features combined. This aligns with theory: slow-moving structural deterioration in affordability is a stronger collapse signal than short-term price momentum. Affordability momentum (the rolling slope of YoY ZHVI growth) ranks seconds, confirming that the *direction and acceleration* of price change matters beyond raw growth rates.

Cross-Table Synthesis:

Table 1’s threshold sensitivity analysis and Table 4’s SHAP importance rankings tell a consistent story: Boise’s earlier onset across every PTI threshold (2016 Q2 at the 4.0x cutoff, versus 2017 Q4 for Austin and Tampa) corresponds with a steeper price_to_income_5yr_chg trajectory in the underlying feature data, reinforcing why this feature carries nearly double the mean absolute SHAP value of the second-ranked affordability_momentum (1.083 vs 0.702). This consistency across an independent sensitivity analysis (Table 1) and a model-internal interpretability method (Table 4) strengthens confidence that the 5-year structural drift in affordability, not short-term price momentum, is the primary mechanism driving collapse. Finding consistent with Malpezzi and Wachter’s (2005) framework distinguishing structural from speculative price deterioration.

Model 2 - Generalization/Scoring Mode (Austin & Boise, 3 Features)

Purpose: Score 371 holdout metros area not in the training set.

Target Variable: is_unaffordable

Features: hpi_yoy_lag, pop_velocity_lag, pop_acceleration_lag

Training Cities: Austin and Boise

Split: 70/30 stratified random split (train : {0: 53, 1:22}; val: {0: 23, 1:10})

Hyperparameters were optimized using Optuna (100 trials, TPE sampler, 5-fold stratified CV, F1 objective), including scale_pos_weight tuning to account for class imbalance.

Table 5: Model 2 Performance (Validation Set)

Metric	Score
Accuracy	78.8%
Precision	71.4%
Recall	50.0%

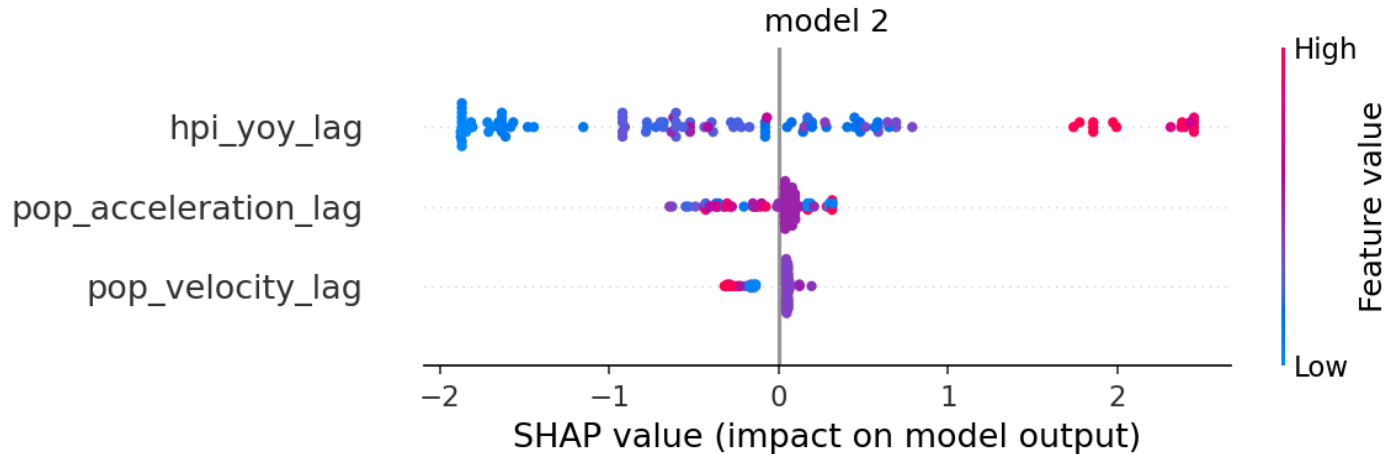
F1-Score	58.8%
ROC-AUC	0.713

The reduced performance of Model 2 relative to model 1 is expected and structurally appropriate. Model 2 uses only 3 broadly available features and is trained on only 2 cities - deliberate trade-off to maximize holdout coverage (19,432 rows, 371 cities), at the cost of precision on the training cities themselves.

Class Imbalance in Results Interpretation: The class imbalance addressed via `scale_pos_tuning` during training carries a direct interpretive consequence. Model 2's 50.0% recall means that roughly half of all quarters that were truly "unaffordable" in the Austin/Boise validation were missing entirely. This is not a simply a method footnote, but it shows how much confidence should be placed in any single holdout metro's `risk_score`, since the same recall ceiling likely applies when scoring the 371 unseen cities. A holdout metro genuinely on a collapse trajectory has a meaningfully higher chance of being under-flagged by Model 2 than by Model 1, reinforcing that Model 2's risk scores should be read as a coarse screening signal, not a precise probability.

Figure 3: SHAP Summary Plot - Model 2

SHAP values were computed for Model 2 using the same Tree Explainer methodology applied to Model 1, revealing a markedly different importance structure than the explanatory model.



Analysis: Unlike Model 1, where `price_to_income_5yr_chg` dominated, Model 2's SHAP beeswarm plot shows `hpi_yoy` as the overwhelming dominant feature, with SHAP values running from approximately -2 to +2.5. A spread roughly 5-8x wider than either population feature. High feature values (pink/red points) for `hpi_yoy_lag` push strongly positive (toward classifying a metro as unaffordable), while low values (blue points) push strongly negative, indicating a clean, monotonic relationship between rapid home price appreciation and collapse risk.

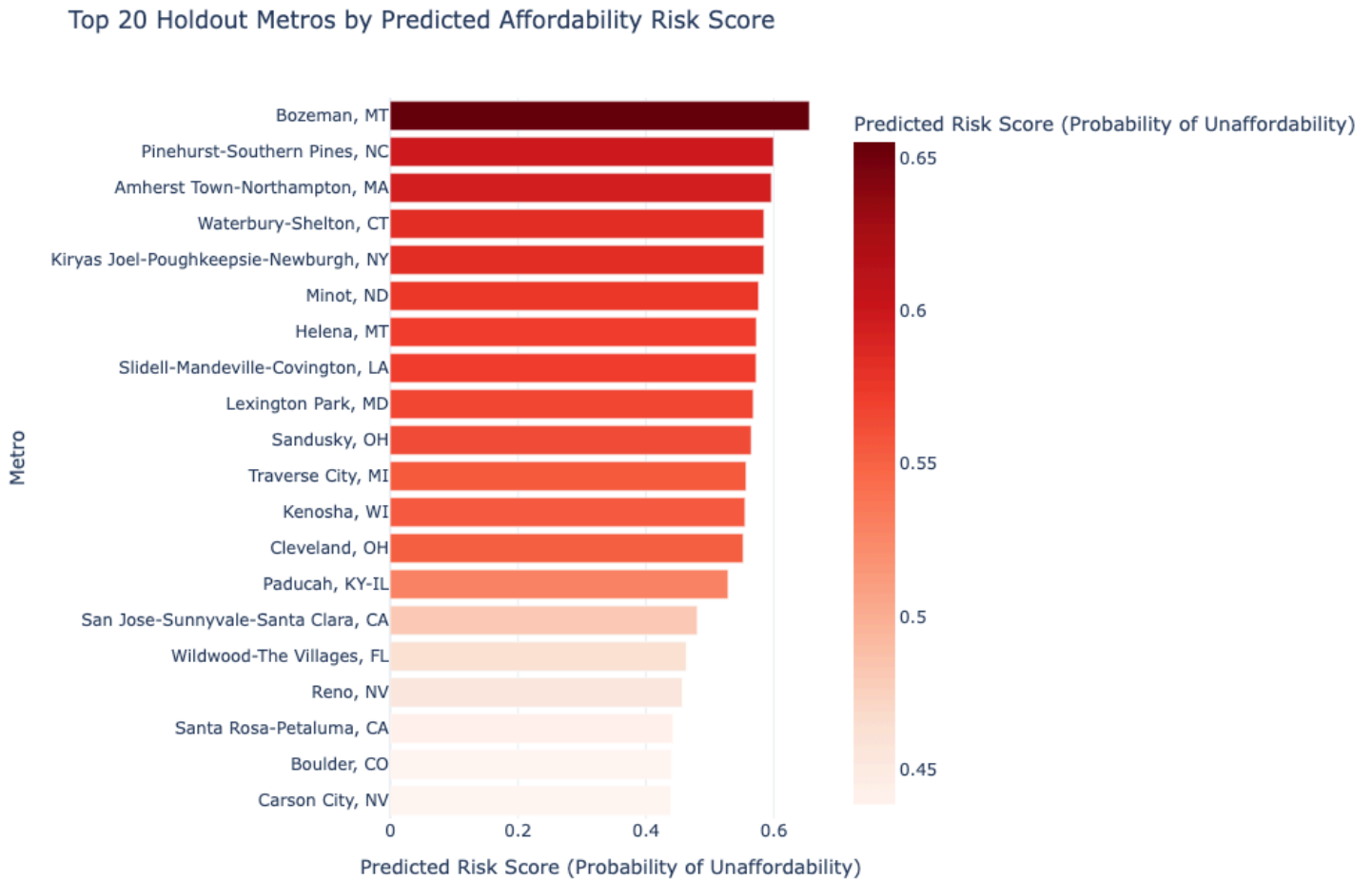
By contrast, `pop_acceleration_lag` and `pop_velocity_lag` show much tighter SHAP value clustering near zero, with only modest spread (roughly -0.5 to +0.5 for acceleration, and even narrower band for velocity). This suggests that while population dynamics contribute some signal, they play a support rather than primary role in Model 2's classification decisions. This is a great contrast to the original hypothesis that population growth would be a strong independent driver, given its established role in the qualitative narrative around Austin's and Boise's in-migration-driven collapse.

This finding has an important implication for the Holdout City Risk Scoring discussed next. The `hpi_yoy_lag` is both the most broadly available feature (66,325 rows across 407 cities) and the dominant driver of Model 2's predictions, the 371-city risk scores effectively being driven primarily by price appreciation momentum rather than population pressure. This helps explain why a small, high-amenity market like Bozeman, MT; which over the past couple of years have experienced extreme home price acceleration for a comparatively smaller population base than Austin or Boise. This proves the model can still rank at the top of the City Risk Index despite structural differences in growth pattern, since the model weights price appreciation far more heavily than demographic velocity.

Holdout City Risk Scoring

Model 2 was applied to all 371 holdout metros to generate a `risk_score` (probability of being unaffordable), averaged across all available quarters per metro. The results are a ranked City Risk Index.

Figure 4: Holdout City Risk Scores



As shown in Figure 4, Bozeman, MT leads the top-15 ranked holdout metros by risk_score, followed by Pinehurst-Southern Pines, NC and Amherst Town-Northampton, MA. The visible drop-off in risk_score after the top few cities suggests a distinct cluster of highest-concern metros rather than a smooth continuum, meaning stakeholders should treat the top 3-5 ranked cities materially priority than the remainder of the top 15.

Note on label absence: Ground-truth collapse_onset labels exist only for Austin, Boise, and Tampa in this dataset. Holdout risk scores represent probabilistic early-warning signals, not validated predictions. This is consistent with the research framing: the goal is prospective flagging, not retrospective accuracy measurement.

What a False Positive or False Negative Means for Stakeholders

Because Model 2's holdout scores cannot be validated against ground-truth labels, the practice stakes of misclassification deserve explicit treatment. A false-positive, meaning a metro incorrectly flagged as high-risk. This carried real economic cost as it could trigger premature rent-control advocacy, deter investment, or cause a regional lender to tighten mortgage underwriting in a market that was never actually on a collapse trajectory, potentially harming the very residents the model aims to protect. A false negative meaning a metro that is genuinely approaching collapse but scores low is worse than a False Positive. This means displacement pressure builds unflagged, and the early-warning system failed at the one job it was built for.

Given Model 2's validated recall of 50.0% (Table 5), stakeholders should treat a "low risk" holdout score with skepticism rather than as reassurance, particularly for metros with rapid recent price appreciation that may simply fall outside the model's detection window. This asymmetry argues for using the City Risk Index as a prioritization tool for further investigation, not as a stand-alone decision input for policy or lending.

Why Bozeman, MT Ranks First in the City Risk Index

Bozeman, Montana emerges as the highest-risk metro in the 371-city holdout scoring, and independent housing market data strongly corroborates this model output. As of early 2024, Bozeman's median home price approached \$1 million, and Gallatin County's 2023 median home price of roughly \$790,000 required an annual household income near \$181,000 to afford. This is nearly double the county's actual median income of approximately \$101,000 implying a price-to-income ratio of ~7.8x. This is well above the project's 5.0x collapse threshold (HCN, 2025; KBZK, 2026).

A 2025 economic report commissioned by the City of Bozeman found that homeowners need to earn 182% of area median income, or ~\$220,000 annually. To afford the median home independently confirming the severity the model's risk_score is designed to detect, even without a verified collapse_onset label for Bozeman in the training data (Montana Free Press, 2026). Montana as a whole ranked the least affordable state for homebuyers in a 2024 analysis, and by mid-2026 reporting placed Montana's statewide home price-to-income ratio above both California's and New York's (Bozeman Daily Chronicle, 2024).

8. Ethics & Limitations

Ethical Considerations

Data source bias: The model is trained exclusively on three Sun Belt metros that became unaffordable during the post-2019 housing surge. All three are relatively young, fast-growing cities with limited historical housing stock constraints. The model may systematically underweight affordability collapse pathways that manifest differently for cities that collapsed prior to 2019.

Threshold subjectivity: The 5.0x PTI threshold, while empirically supported and consistent with existing housing literature, is still a human-defined cutoff. Some metros may experience functional displacement at 4.5x (e.g., metros with lower-income distributions), while may remain stable at 5.5x. The threshold sensitivity analysis (Table 1) documents how results shift under alternative thresholds.

Proxy labeling: The project uses $PTI > 5.0$ as a *proxy* for affordability collapse. Actual collapse - displacement, community disruption, municipal budget strain is more complex and dimensional than any single numeric threshold. The label is a tractable operationalization, not a comprehensive definition.

A “high-risk” label can itself move markets: Because this project’s output is a ranked, publicly interpretable risk score, there is a real possibility that labeling a metro “high-risk” becomes a self-fulfilling or self-defeating signal. If investors treat the label as a buy signal ahead of anticipated appreciation, or if it discourages needed development in a market that was misclassified. This risk is inherent to a predictive tool that names specific places, and it argues for presenting the City Risk Index as a screening prioritization aid for further human review. **NOT AS A STANDALONE PUBLIC-FACING RANKING.**

Limitations:

Limitations	Description
Small training set	Only 3 cities have a ground-truth collapse label. Model 1 trains on ~126 rows.
Tampa population gap	Tampa lacks population data in the pipeline, preventing use of <code>pop_velocity_lag</code> in the full 3-city model.
No holdout labels	The 371 holdout metros have no verified collapse labels; risk scores cannot be externally validated against known outcomes.
Temporal coverage	ZHVI data begins ~2010; features requiring long lags (e.g., 5-year PTI change) are unavailable for the earliest quarters.
Macro confounders	The 2020-2022 housing surge was driven partly by COVID-specific factors (remote work, historically low mortgage rates) that may not recur. The model may over-fit to a specific macro regime.
CBSA mismatch	Tampa requires a manual CBSA crosswalk correction (45300 -> 45294), and any similar mismatches in holdout cities could introduce silent join failures.
Geographic training bias in holdout rankings	All three training cities are Sun Belt/mountain-west metros. Risk scores for structurally different regions. Specifically those in the Rust Belt and Northeast metros, which may face affordability stress through slow-growth pathways rather than rapid price appreciation. This should be treated with reduced confidence, since the model was never trained on that collapse pattern. Any Rust Belt or Northeast metro appearing in the top-15 holdout

	lists warrants additional qualitative scrutiny before being treated as a comparably reliable to a Sun Belt or mountain-west match like Bozeman.
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9. Conclusion

This project demonstrates the affordability collapse in mid-size U.S. metros is detectable with high accuracy using a small number of lagged, publicly available housing and economic indicators. The **5-year change in price-to-income ratio** is the single most powerful early-warning feature. Outperforming short-term momentum and HPI growth individually. This 0.943 ROC-AUC of the explanatory model confirms that the three training cities (Austin, Boise, Tampa) exhibit structurally consistent pre-collapse signal patterns that can be learned and generalized.

Directly answering this project's research question: price-to-income 5-year change, affordability momentum, and HPI year-over-year growth are the socioeconomic and housing market indicators that best predict affordability collapse onset in this sample dataset. Model 1 demonstrates these indicators and can identify at-risk with strong accuracy (90.7%) among cities structurally similar to the training set. Model 2 demonstrates these same indicators, in reduced form and can extend the identification to a much broader set of metros. With the explicit downside that the confidence is scaled down for cities structurally dissimilar to Austin, Boise, and Tampa.

The two-model architecture; one for interpretability, one for broad scoring balances both the explanatory depth with geographic coverage. The holdout scoring pipeline applied to 371 metros produces a deployable City Risk Index that city planners, regional banks, and housing advocates can use as a forward-looking resource. The Bozeman, MT case study demonstrates this index practical value: despite lacking a ground-truth label, the mode's top ranked risk score aligns closely with independently reported, real-world affordability distress in that market. This suggests the City Risk index has a genuine early-warning utility beyond the three training cities, provided its false-negative risk and geographic training bias are kept in view by any stakeholder using it.

Future work should prioritize:

1. Expanding training labels to additional collapsed metros (e.g., Nashville, Phoenix, Denver, and at least one Rust Belt or Northeast metro to test cross-regional generalization).
2. Resolving Tampa's data gap
3. Incorporating permit issuance data as a supply-side feature
4. Building an R Shiny or Streamlit dashboard for interactive risk exploration

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Appendix

Appendix A: Key Variables Definitions

Variable	Definition
price_to_income_ratio	Quarterly median ZHVI / annual ACS median household income
collapse_onset	First quarter where PTI first exceeds 5.0 after a period below 5.0
is_unaffordable	Binary flag: $PTI > 5.0$
hpi_3yr_chg	$(HPI_t / HPI_{\{t-12\}}) - 1$; 3-year price appreciation
zhvi_yoy	$(ZHVI_t / ZHVI_{\{t-4\}}) - 1$; year-over-year home value growth
affordability_momentum	12-quarter rolling OLS slope of lagged zhvi_yoy
price_to_income_5yr_chg	$PTI_{lag} - PTI_{lag_20}$; 5-year structural drift in affordability
pop_velocity_lag	First difference of annual population YoY growth, lagged 4 quarters

Appendix B: Model Architecture Summary

	Model 1	Model 2
Purpose	Explanatory / SHAP	Generalization / Scoring
Training cities	Austin, Boise, Tampa	Austin, Boise
Features	4 (ZHVI + HPI + PTI)	3 (HPI + Population)
Train rows	126	75
Validation rows	54	33
ROC-AUC	0.943	0.713
Holdout cities	N/A	371